$$= (\vec{p}\cdot\vec{d})\vec{c} - (\vec{p}\cdot\vec{c})\vec{d}$$

$$= ((\vec{a}\times\vec{b})\cdot\vec{d})\vec{c} - ((\vec{a}\times\vec{b})\cdot\vec{c})\vec{d} = [\vec{a},\vec{b},\vec{d}]\vec{c} - [\vec{a},\vec{b},\vec{c}]\vec{d}$$

Similarly, taking $\vec{q} = \vec{c}\times\vec{d}$, we get

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = (\vec{a}\times\vec{b})\times\vec{q}$$

$$= (\vec{a}\cdot\vec{q})\vec{b} - (\vec{b}\cdot\vec{q})\vec{a}$$

$$= [\vec{a},\vec{c},\vec{d}]\vec{b} - [\vec{b},\vec{c},\vec{d}]\vec{a}$$

Example 6.22

If $\vec{a} = -2\hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{b} = 3\hat{i} - \hat{j} + 3\hat{k}$, $\vec{c} = 2\hat{i} - 5\hat{j} + \hat{k}$, find $(\vec{a}\times\vec{b})\times\vec{c}$ and $\vec{a}\times(\vec{b}\times\vec{c})$. State whether they are equal.

Solution

$$\text{By definition, } \vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & -2 \\ 3 & -1 & 3 \end{vmatrix} = 7\hat{i} - 7\hat{k}\ .$$

$$\text{Then, } (\vec{a}\times\vec{b})\times\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 0 & -7 \\ 2 & -5 & 1 \end{vmatrix} = -35\hat{i} - 21\hat{j} - 35\hat{k}\ . \quad \text{... (1)}$$

$$\vec{b}\times\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 3 \\ 2 & -5 & 1 \end{vmatrix} = 14\hat{i} + 3\hat{j} - 13\hat{k}\ .$$

$$\text{Next, } \vec{a}\times(\vec{b}\times\vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & -2 \\ 14 & 3 & -13 \end{vmatrix} = -33\hat{i} - 54\hat{j} - 48\hat{k}\ . \quad \text{... (2)}$$

Therefore, equations (1) and (2) lead to $(\vec{a}\times\vec{b})\times\vec{c} \neq \vec{a}\times(\vec{b}\times\vec{c})$.

Example 6.23

If $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} - \hat{j} - 4\hat{k}$, $\vec{c} = 3\hat{j} - \hat{k}$ and $\vec{d} = 2\hat{i} + 5\hat{j} + \hat{k}$, verify that

(i) $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = [\vec{a},\vec{b},\vec{d}]\vec{c} - [\vec{a},\vec{b},\vec{c}]\vec{d}$

(ii) $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = [\vec{a},\vec{c},\vec{d}]\vec{b} - [\vec{b},\vec{c},\vec{d}]\vec{a}$

Solution (i)

By definition,

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 1 & -1 & -4 \end{vmatrix} = 4\hat{i} + 4\hat{j},\ \vec{c}\times\vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3 & -1 \\ 2 & 5 & 1 \end{vmatrix} = 8\hat{i} - 2\hat{j} - 6\hat{k}$$

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